8. **Simplify and express the result in power notation:**

a. $\left(\frac{5}{2}\right)^6 \times \left(\frac{5}{2}\right)^2$ b. $\left(\frac{11}{3}\right)^{11} \times \left(\frac{11}{3}\right)^2$ c. $\left(\frac{3}{4}\right)^7 \div \left(\frac{3}{4}\right)^4$ d. $\left\{\frac{7}{10}^{-2}\right\}^6$

e. $\left(\frac{4}{5}\right)^3 \div \left(\frac{4}{5}\right)^8$ f. $\left[\left(\frac{2}{3}\right)^4\right]^2$ g. $\left[\left(\frac{-3}{4}\right)^3\right]^4$ h. $\left(\frac{1}{9^3}\right)^4$

9. **Simplify :**

a. $(3^2 - 2^2) \times \left(\frac{2}{3}\right)^{-3}$ b. $\left\{\left(\frac{1}{2}\right)^{-1} \times (-4)^{-1}\right\}^{-1}$ c. $\left[\left\{\left(\frac{-1}{4}\right)^2\right\}^{-2}\right]^{-1}$

d. $\left\{\left(\frac{1}{3}\right)^{-3} - \left(\frac{1}{2}\right)^{-3}\right\} \div \left(\frac{1}{4}\right)^{-3}$ e. $\left\{\left(\frac{2}{3}\right)^2\right\}^3 \times \left(\frac{1}{3}\right)^{-4} \times 3^{-1} \times 6^{-1}$ f. $\left\{\left(\frac{9}{10}\right)^2\right\}^{-3} \times \left(\frac{9}{10}\right)^4 \times 3^{-1} \times \frac{1}{12}$

10. **Find the value of x in each of the following :**

a. $\left(\frac{5}{3}\right)^{-5} \times \left(\frac{5}{3}\right)^{-11} = \left(\frac{5}{3}\right)^{8x}$ b. $\left(\frac{2}{9}\right)^3 \times \left(\frac{2}{9}\right)^{-6} = \left(\frac{2}{9}\right)^{2x-1}$ c. $\left(\frac{-1}{2}\right)^{-19} \div \left(\frac{-1}{2}\right)^8 = \left(\frac{-1}{2}\right)^{-2x+1}$ d. $\left(\frac{3}{2}\right)^{-3} \times \left(\frac{3}{2}\right)^5 = \left(\frac{3}{2}\right)^{2x+1}$

11. **If $\frac{p}{q} = \left(\frac{3}{2}\right)^{-2} \div \left(\frac{6}{7}\right)^0$, then find the value of $\left(\frac{p}{q}\right)^{-3}$.**

12. **If $x = \left(\frac{4}{5}\right)^{-2} \div \left(\frac{1}{4}\right)^2$, then find the value of x^{-1}.**

13. **Evaluate :**

a. $\frac{2^5 \times 3^4 \times 25}{81 \times 5^2}$ b. $(3^2)^2 \quad (\quad)^2 \div (27)^3$ c. $\frac{\left(\frac{5}{7}\right)^3 \times \frac{35}{250}}{\left(\frac{-7}{5}\right)^2 \div \frac{25}{14}}$

14. **By what number should we multiply 3^{-7} so that the product may be equal to 3?**

15. **By what number should $(-4)^5$ be divided so that the quotient may be equal to 4^{-2}?**

16. **By what number should $(-8)^{-1}$ be multiplied so that the product may be equal to $(10)^{-1}$?**

17. **By what number should $(-24)^{-1}$ be divided so that the quotient may be $(3)^{-1}$?**

18. **What number should be multiplied with $(5)^{-3}$ so that the product becomes 125?**

19. **By what number should $\left(\frac{3}{8}\right)^{-3}$ be divided to get $\left(\frac{9}{64}\right)$?**

20. **By what number should $\left(\frac{4}{7}\right)^{-2}$ be divided so that the quotient becomes $\left(\frac{7}{16}\right)$?**

USE OF EXPONENTS TO EXPRESS SMALL NUMBERS IN STANDARD FORM

In the previous class, we have learnt how to express very large numbers in standard form by using exponents of 10. Let us now see how can we write very small numbers in standard form.

Remember the following points while doing so:

- In the standard form, there is only one digit to the left of the decimal point.
- For a number greater than or equal to 10, the exponent will always be positive.

Example: a. 65931000 can be written in the standard form as 6.5931×10^7.

b. 42,00,00,000 can be written in the standard form as 4.2×10^8.

For a number smaller than 1, the exponent will always be negative.

Example: a. $0.00006 = \frac{6}{100000} = \frac{6}{10^5} = 6 \times 10^{-5}$ b. $0.000485 = 4.85 \times 10^{-4}$

[Since the number has to be between 1 and 10, we move decimal point 4 places to the right, hence the p of 10 is –4.]

- In case a number is greater than or equal to 1 and less than 10, the exponent will always be 0.

Example: 8 can be written in the standard form as 8.0×10^0.

Solved Examples

Example 12: Write the following numbers in standard form:

a. 0.6573 b. 0.000005 c. 0.00000025 d. 0.0000031

Solution:

a. $0.6573 = 6.573 \times 10^{-1}$ [$\because$ The decimal point is moved 1 place to the rig

b. $0.000005 = 5 \times 10^{-6}$ [$\because$ The decimal point is moved 6 places to the rig

c. $0.000000254 = 2.54 \times 10^{-7}$ [$\because$ The decimal point is moved 7 places to the rig

d. $0.0000031 = 3.1 \times 10^{-6}$ [$\because$ The decimal point is moved 6 places to the rig

Example 13: Write the following numbers in usual form:

a. 6.57×10^4 b. 9.24×10^{-5} c. 2×10^{-6}

Solution:

a. $6.57 \times 10^4 = 6.57 \times 10{,}000 = 65{,}700$

b. $9.24 \times 10^{-5} = 0.0000924$

c. $2 \times 10^{-6} = 0.000002$

Example 14: The diameter of the Sun is 1.4×10^9 m and that of the Earth is 1.2756×10^7 m. Approximately, how many times is the diameter of the Sun greater as compared to that of the Earth?

Solution: Diameter of the Sun $= 1.4 \times 10^9$ m

Diameter of the Earth $= 1.2756 \times 10^7$ m

$$\therefore \frac{\text{Diameter of the Sun}}{\text{Diameter of the Earth}} = \frac{1.4 \times 10^9}{1.2756 \times 10^7} = \frac{1.4 \times 10^2 \times 10^7}{1.2756 \times 10^7} = \frac{1.4 \times 10^2}{1.2756}$$

$$= \frac{1.4}{1.3} \times 100 = 1 \times 100 = 100 \qquad [\because 1.2756 = 1.3] \; [\because \frac{1.4}{1.3} = 1]$$

Thus, the diameter of the Sun is approximately 100 times the diameter of the Earth.

Example 15: The distance between Sun and Earth is 1.496×10^{11}m and the distance between Earth and Moo 3.84×10^8 m. During solar eclipse, moon comes in between Earth and Sun. At that time, what is distance between Moon and Sun?

Solution : Distance between the Sun and the Earth $= 1.496 \times 10^{11}$ m.

Distance between the Earth and the Moon $= 3.84 \times 10^8$ m.

Distance between the Sun and the Moon $= (1.496 \times 10^{11} - 3.84 \times 10^8)$ m.

$= (1.496 \times 10^3 \times 10^8 - 3.84 \times 10^8)$ m

$= (1.496 \times 1000 - 3.84) \times 10^8$ m

$= (1496 - 3.84) \times 10^8$ m $= 1492.16 \times 10^8$ m

Thus, the distance between the Sun and the Moon is 1492.16×10^8 m.

Remember

To add or subtract numb in the standard fo convert them into numb with the same exponen

EXERCISE 2(B)

1. **Express the following numbers in standard form:**

a. 0.0000000000086 b. 0.00000000000943 c. 8020000000000000 d. 0.52

e. 0.00000000839 f. 32860000000 g. 0.000014356 h. 0.000907

2. **Express the following numbers in usual form:**

a. 3.05×10^{-6} b. 4.6×10^4 c. 8×10^{-8} d. 2.0001×10^9

e. 5.8×10^{12} f. 5.61492×10^6 g. 1.89×10^{-5} h. 3.14271×10^5

Express the numbers appearing in the following statements in standard form:

a. 1 micron is equal to $\frac{1}{10{,}00{,}000}$ m.

b. Charge of an electron is 0.000,000,000,000,000,000, 16 coulomb.

c. Size of a bacteria is 0.0000005 m.

d. Size of a plant cell is 0.00001275 m.

e. Thickness of a thick paper is 0.07 mm.

f. The speed of light is 300,000,000 metre per second.

g. A polio Virus has as diameter of 0.000000012 m.

h. The Volume of the Earth is 1,083,230,000,000 cubic km.

4. The size of a red blood cell is 0.000007 m and the size of a plant cell is 0.00001275 m. Compare these two size by writing them in the standard form.

5. The mass of the Earth is 5.976×10^{24} kg and the mass of the sun is 1.98892×10^{30} kg. Find the sum of their masses.

Life Skills and Values

Aarav saves ₹5 in the first week of January 2024, ₹15 in the second week, ₹45 in the third week, and so on, tripling the amount each week. How much will Aarav have saved by the end of the second week of February? What quality does Aarav demonstrate through this habit of saving? How frequently do you save money? Can you think of different ways to save money? What value is emphasised by Aarav's consistent saving habit?

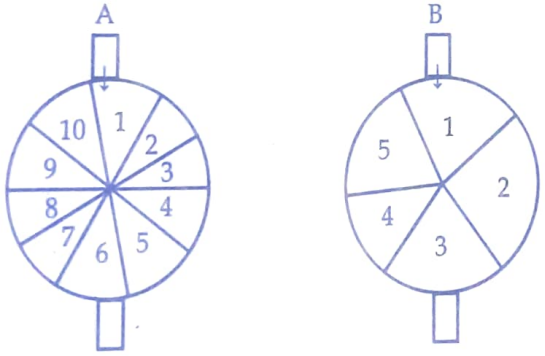
Subject Integration

Research the total number of satellites orbiting the Earth. Additionally, determine the average distance between Earth and these satellites. Express the distance in scientific notation for better clarity.

Maths Lab ACTIVITY

Objective: To enhance the knowledge of exponents.

Material Required: Two spinners: Base spinner (A) and Exponent spinner (B) as shown below.



Base Spinner Exponent Spinner

Steps:

Divide the class into groups of two students each. There will be three rounds in the game.

Round I

1. The first students spins the spinner A to get the base (a) and the spinner B to get the exponent (m). The first exponential becomes (a^m). He again spins the spinner B to get the second exponential (a^n). The second student performs the same steps.
2. Each student finds the value of $a^m \times a^n$.
3. Each students finds the value of $a^m \div a^n$.
4. Add the values obtained in Step 2 and Step 3.
5. The student who gets the maximum sum is the winner.

Round II

1. Repeat Step 1 of Round I.
2. Each students finds the value of $a^m \times a^n$.
3. Each students find the values of $(a^m)^n$.
4. Find the difference between the values obtained in Step 2 and Step 3.
5. The student who gets the minimum difference is the winner.

Round III

1. Repeat Step 1 of Round I.
2. Each student finds the value of $a^m \div a^n$.
3. Each students find the values of $(a^m)^n$.
4. Find the difference between the values obtained in Step 2 and Step 3.
5. The student who gets the minimum difference is the winner.

The students who wins the most number of rounds is declared to be the final winner.

Higher Order Thinking Skills

2. Simplify: a. $\frac{1}{1+x^{q-p}}+\frac{1}{1+x^{p-q}}$ b. $\frac{(a^{p+q})^2\times(a^{q+r})^2\times(a^{r+p})^2}{(a^p.\,a^q.\,a^r)^3}$

3. The size of a red blood cell is 0.000007 m and that of a plant cell is 0.00001275 m. Approximately, how m times is the size of a plant cell large than the size of a red blood cell?

4. A spacecraft is travelling at a speed of 12 km/sec. How much time will it take to reach the surface of Moon, if the distance between the Moon and the Earth is 3.48×10^8 m?

Rewind Yourself

1. **Simplify and express as rational numbers with a positive exponent:**

a. $\left(\frac{-5}{17}\right)^{-2}$ b. $(15)^{-3}$ c. $\left(\frac{7}{2}\right)^3\times\left(\frac{7}{2}\right)^2$ d. $\left(\frac{-11}{3}\right)^{-4}\times\left(\frac{-11}{3}\right)^{-3}$

e. $\left(\frac{2}{5}\right)^{-4}\div\left(\frac{2}{5}\right)^2$ f. $\left(\frac{5}{12}\right)^{-2}\div\left(\frac{5}{12}\right)^{-5}$ g. $\left[\left(\frac{6}{19}\right)^{-3}\right]^4$ h. $\left(\frac{3}{4}\right)^0$

2. **Simplify:**

a. $(-4)^9\div(-4)^3$ b. $8^{13}\div8^{19}$ c. $2^{-3}\times2^4$ d. $(3^3)^4\times3^3\div3^7$

e. $2^{\frac{3}{2}}\times4^{-\frac{1}{2}}$ f. $(144)^{-\frac{1}{2}}$ g. $\{(2)^{-2}\times(2)^{-2}\}^2$ h. $\frac{3^{-5}\times10^{-5}\times81}{3^{-4}\times30^{-5}}$

3. **Simplify:**

a. $\left(\frac{3}{4}\right)^2\times\left(\frac{3}{4}\right)^{-2}\div2^{-2}$ b. $\left(\frac{1}{3}\right)^5\div\left(\frac{1}{3}\right)^4\div\frac{1}{3}$ c. $(-3)^4\times\left(\frac{5}{3}\right)^4$ d. $\left[\left(\frac{2}{3}\right)^2\right]^3\times\left(\frac{1}{3}\right)^{-2}\times3^{-1}\times\frac{1}{6}$

4. **Show that:**

a. $\left(\frac{9}{13}\times\frac{-11}{17}\right)^{-8}=\left(\frac{13}{9}\right)^8\times\left(\frac{17}{-11}\right)^8$ b. $\left[\left(\frac{-12}{19}\right)\times\left(\frac{-27}{43}\right)\right]^{-7}=\left(\frac{19}{12}\right)^7\times\left(\frac{43}{27}\right)^7$

5. **Find x so that $\left(\frac{5}{3}\right)^{-5}\times\left(\frac{5}{3}\right)^{-11}=\left(\frac{5}{3}\right)^{8x}$.**

6. **Find m so that $\left(\frac{2}{9}\right)^3\times\left(\frac{2}{9}\right)^{-6}=\left(\frac{2}{9}\right)^{2m-1}$.**

7. **Simplify:**

a. $\frac{8^{-1}\times5^3}{2^{-4}}$ b. $(5^{-1}\times2^{-1})\times6^{-1}$ c. $\left(\frac{-1}{3}\right)^4\div\left(\frac{-1}{3}\right)^8\times\left(\frac{-1}{3}\right)^5$

d. $\frac{x^{m+n}\times x^{n+l}\times x^{l+m}}{(x^m\times x^n\times x^l)^2}$ e. $\frac{a^{5+m}\times(a^2)^{3m+1}}{a^{7m-2}}$ f. $\frac{a^{2p+1}\times a^{(2p+1)(2p-1)}}{a^{p(4p-1)}\times(a^2)^{2p+3}}$

8. **Show that:**

a. $\left(\frac{x^a}{x^b}\right)^{a-b}\times\left(\frac{x^b}{x^c}\right)^{b-c}\times\left(\frac{x^c}{x^a}\right)^{c-a}=1$ b. $(a+b)^{-1}(a^{-1}+b^{-1})=(ab)^{-1}$

9. **Solve the following equations:**

a. $3^{3x-5}=\frac{1}{9^x}$ b. $2^{2x+2}=4^{2x-1}$ c. $3^{2x-1}=(\sqrt{27})^{4x-1}$ d. $\left(\frac{2}{3}\right)^x\times\left(\frac{2}{3}\right)^{x+1}=\left(\frac{2}{3}\right)^5$

10 **Prove that:**

a. $\frac{x^a}{x^b}\times\frac{x^b}{x^c}\times\frac{x^c}{x^a}=1$ b. $\left(\frac{a^l}{a^m}\right)^{l+m}\times\left(\frac{a^m}{a^n}\right)^{m+n}\times\left(\frac{a^n}{a^l}\right)^{n+l}=1$

11. **By what number, $(-8)^{-1}$ should be multiplied so that the product may be equal to $(-6)^{-1}$?**

12. **By what number, $(-18)^{-1}$ should be divided so that the quotient may be 6^{-1}?**

13. **Write the following numbers using standard form:**

a. 0.000000000081 b. 63,80,00,000 c. 0.0000014365 d. 36,25,00,00,000

14. Write the following numbers in the usual form:

a. 1.79×10^{-5} b. 5.6×10^{-9} c. 3.14271×10^{6} d. 5.697×10^{11}

15. The diameter of Jupiter is 1.42984×10^{8} m and the diameter of Pluto is 2.32×10^{6} m. Find the difference between their diameters.

Multiple Choice QUESTIONS

Tick (✓) the correct answer.

1. The value of $\frac{2^{2001} + 2^{1999}}{2^{2000} - 2^{1999}}$ is _____.

a. 2 ☐ b. 5 ☐ c. $2^{1000} + 1$ ☐ d. 10 ☐

2. When simplified, $(x^{-1} + y^{-1})$ is equal to _____.

a. $x + y$ ☐ b. $\frac{x + y}{xy}$ ☐ c. xy ☐ d. $\frac{1}{xy}$ ☐

3. The value of $x + x(x^{x})$, when $x = 2$ is _____.

a. 10 ☐ b. 16 ☐ c. 18 ☐ d. 36 ☐

4. Multiplicative inverse of 10^{-100} is _____.

a. 10^{-100} ☐ b $\frac{1}{10^{-100}}$ ☐ c. 10^{100} ☐ d. $(10^{-50})^{3}$ ☐

5. The value of $(3^{2})^{3} + \left(\frac{2}{3}\right)^{0} + 3^{5}$ is _____.

a. 930 ☐ b. 973 ☐ c. 932 ☐ d. 950 ☐

6. If $\frac{10}{3} \times 3^{x} - 3^{x-1} = 81$, then the value of x is _____.

a. 2 ☐ b. 1 ☐ c. 3 ☐ d. 0 ☐

7. The value of expression $(8^{0} - 3^{0}) \times (8^{0} + 3^{0})$ is equal to _____.

a. 0 ☐ b. 1 ☐ c. 2 ☐ d. 3 ☐

8. The reciprocal of the rational number $\left(\frac{3}{2}\right)^{-2} \div \left(\frac{1}{3}\right)^{-3}$ is _____.

a. $\frac{240}{4}$ ☐ b. $\frac{241}{4}$ ☐ c. $\frac{243}{4}$ ☐ d. $\frac{241}{4}$ ☐

9. The value of $(3^{0} - 4^{0}) \times 5^{2}$ is _____.

a. 25 ☐ b. 0 ☐ c. −25 ☐ d. 1 ☐

10. The value of x in exponential equation $2^{x-4} = 1$ is equal to _____.

a. 4 ☐ b. 3 ☐ c. 2 ☐ d. 1 ☐

11. The standard form of 0.00000037 is _____.

a. 3.7×10^{-6} ☐ b. 3.7×10^{-7} ☐ c. 3.7×10^{-5} ☐ d. 3.7 ☐

12. If $\left\{\left(\frac{1}{3}\right)^{-1} - \left(\frac{1}{4}\right)^{-1}\right\}^{m} = 1$, then the least positive integral value of m is _____.

a. 0 ☐ b. 2 ☐ c. 1 ☐ d. 3 ☐